

Daniel Aarao Reis Arturi

Research Scientist and Research Engineer

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RESEARCH PROFILE

Research scientist who develops rigorous experiments, reproducible evidence, and working systems for open-ended problems. Current work focuses on AI for mathematics and mechanistic interpretability, supported by research in database systems, quantum optimization, and sustainable machine learning.

Research interests AI for mathematics, human-controlled formalization, mechanistic interpretability, research systems, and reproducible evaluation

RESEARCH EXPERIENCE

NYU Tandon School of Engineering, VIDA Center

May 2026-Present

Research Scientist | DARPA expMath | [Lea](#) | [Program](#)

- Helped develop Lea, an open-source platform for mathematician-controlled formalization that supports proof decomposition, mid-run intervention, trajectory inspection, manual correction, and auditable Lean 4 output.
- Worked on the agentic harness and its research infrastructure, integrating project-aware execution, configurable sub-agent roles, Lean checking, typed event streams, persistent proof history, and shared state across standalone and Overleaf clients.
- Built the React/Vite/TypeScript and FastAPI research interface and developed the Overleaf-to-Lean workflow, including dependency-aware formalization, counterexample handling, self-repair, theorem-specific review, and GitHub export.
- Contributed to long-horizon formalization evaluations, related-work synthesis, architecture documentation, case-study analysis, and the human-in-the-loop research framing; GitHub listed me as Lea's leading public contributor with 144 contributions in September 2026.

Algoverse AI Research

Apr 2025-Present

Research Intern | AI Safety and Mechanistic Interpretability | [Paper](#) | [Project](#) | [Code](#)

- Studied emergent misalignment through the parameter geometry of LoRA-adapted Qwen2.5-7B and Llama 3.1-8B models. Across harmful tasks, mean principal angles were about 20 degrees versus an approximately 85-degree random baseline, with projection overlap near 0.8.
- Used linear mode connectivity to test functional equivalence across independently fine-tuned models; cross-task interpolations retained misaligned behavior while preserving more than 90% coherence. Extended the work to post-hoc and during-training interventions and GSM8K retention.
- Co-authored the NeurIPS 2025 workshop paper as a joint first author and began a separate study of how multi-thousand-turn conversations and persistent memory affect model persona stability.

McGill University and University of Passau

Aug 2025-Mar 2026

Research Intern | Graph Database Systems | [Preprint](#) | [Artifact](#)

- Adapted Dewey and PrePost structural indexes to tree-shaped property-graph data and evaluated baseline traversal and indexed variants across Apache AGE, Kuzu, and Neo4j using 100,000-node synthetic trees and LDBC SNB workloads.
- Measured 6.7x to more than 300,000x microbenchmark speedups and 1.1x to 16x gains on three LDBC queries in AGE, while documenting mixed results in native graph engines and index-maintenance costs; released a Dockerized reproducibility artifact and first-authored the EDBT 2027 submission.

McGill University, Distributed Information Systems Group

Jan-Jun 2026

Research Intern | Quantum Computing for Data Systems | [Paper](#) | [Artifact](#)

- Formulated fault-tolerant distributed data allocation through binary-slack, calibrated unbalanced-penalty, and domain-wall QUBO encodings, with exact ILP and brute-force oracles for feasibility and optimality checks.
- Evaluated D-Wave simulated annealing, hybrid CQM, Advantage hardware, and QAOA while separating QPU time from end-to-end cloud overhead. The study found no quantum advantage, mapped current scaling barriers, and produced a first-author paper presented at VLDB 2026 QC&DKM.

PUBLICATIONS AND MANUSCRIPTS

Daniel Aarao Reis Arturi*, Eric Zhang*, Andrew Ansah, Kevin Zhu, Ashwinee Panda, and Aishwarya Balwani. "[Shared Parameter Subspaces and Cross-Task Linearity in Emergently Misaligned Behavior](#)". arXiv:2511.02022, 2025. Joint first author. [[Project](#)] [[Code](#)]

* Joint first authors with equal contributions; authors listed alphabetically. Expanded manuscript: Of Basins and Vaccines: Emergent Misalignment Through the Lens of Parametric Geometry (ICLR submission).

Daniel Aarao Reis Arturi, Emilie Fahber, Stefanie Scherzinger, and Bettina Kemme. "[Evaluating a Quantum Solution: How hard can it be?](#)". 2nd Workshop on Quantum Computing and Data/Knowledge Management, VLDB 2026. First author and presenter. [[Artifact](#)]

Experience report and reproducibility artifact covering QUBO formulation, simulators, hybrid solvers, D-Wave Advantage hardware, QAOA, classical baselines, and practical scaling limits.

Daniel Aarao Reis Arturi, Christoph Köhnen, George Fletcher, Bettina Kemme, and Stefanie Scherzinger. "[Seeing the Trees for the Forest: Leveraging Tree-Shaped Substructures in Property Graphs](#)". arXiv:2603.12476, 2026. Under review at EDBT 2027. [[Artifact](#)]

First-author vision paper with an open, cross-system reproducibility artifact. Selected for the 2026 McGill Science Undergraduate Research Showcase.

SELECTED RESEARCH SYSTEMS AND STUDIES

Profiling and Optimising Energy Efficiency in GFlowNet Molecular Generation | *Python, PyTorch, CUDA, CodeCarbon, Slurm* [[Code](#)] [[Report](#)]

- Instrumented a ~600M-parameter GraphTransformerGFN with timing-only, CodeCarbon, and CUDA-synchronized measurement modes at less than 0.2% overhead; found molecular sampling consumed 94.4% of step time while GPU utilization remained below 13%.
- Parallel sampling with four workers improved throughput about 1.9x and reduced total energy use by approximately 47%; automated Slurm sweeps covered batch sizes 64, 128, and 256 across three seeds and all measurement modes.

NYC Housing Research | *Python, FastAPI, SQLite, retrieval and evaluation* [[Code](#)]

- Built a local-first cited research system over official NYC legal sources, separating keyless citation search from optional semantic indexing and evidence-cited generation while preserving source versions, provenance, explicit coverage gaps, and per-resource provider consent.
- Developed retrieval and answer-evaluation infrastructure, citation-support checks, report-bound human-review workflows, explicit provider-cost approvals, and documented gates separating software regression evidence from substantive housing-law review.

EDUCATION

New York University, Courant Institute of Mathematical Sciences | M.S. Computer Science | Fall 2026-Present

McGill University | B.Sc. Honours Software Engineering | May 2026 | GPA 3.99/4.0 | First Class Honours

Selected coursework Graduate Distributed Systems, Graduate Applied ML, Graduate Computer Networks, Sustainability for ML, Algorithms, Operating Systems, Database Systems, Probability and Statistics

AWARDS AND RECOGNITION

NeurIPS 2025 workshops UniReps - Oral Award | MechInterp - Honourable Mention | CogInterp - Poster Acceptance

McGill Science Undergraduate Research Showcase (2026) | Wing Hing Chan Scholarship in Science | Algoverse Merit Scholarship (twice) | McGill Dean's Honour List

RESEARCH METHODS AND TECHNICAL SKILLS

Methods Experimental design, reproducible evaluation, retrieval and citation evaluation, mechanistic interpretability, parameter-subspace analysis, linear mode connectivity, QUBO, statistical analysis, technical writing

Research stack Python, PyTorch, NumPy, pandas, scikit-learn, Jupyter, CUDA, Slurm, CodeCarbon, D-Wave Ocean, Qiskit Aer

Systems Lean 4, Mathlib, FastAPI, React, TypeScript, SQLite, PostgreSQL, Docker, Git, Apache AGE, Kuzu, Neo4j